HU Collection

Test of Mathematics for University Admission

Mock Set B Paper 2

Mathematical Reasoning

Time allowed:	**75 minutes**
Questions:	20 multiple choice
Calculators:	not permitted
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Date written:	19 September 2026

Name: _____

Date sat: _____

Score: _____ / 20

Instructions

Answer all 20 questions. Each question has exactly one correct option and is worth one mark. No marks are deducted for wrong answers. Unless a question says otherwise, all angles are in radians. Throughout, "must be true" means true in every case allowed by the given information, and "can be true" means true in at least one such case. Record your answers as letters on the grid below before consulting the solutions.

Difficulty note. This paper tests reasoning rather than technique, at a level above the standard TMUA. The ideas are the ones a first olympiad course teaches: find the quantity that does not change, count something two ways, look at remainders, find the losing positions. None of the questions is a known problem with a known answer; each is a fresh instance, and each has at least one option that a plausible half-argument makes look correct.

Q	1	2	3	4	5	6	7	8	9	10
Ans										
Q	11	12	13	14	15	16	17	18	19	20
Ans										

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Question 1

The twenty numbers $1, \frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{20}$ are written on a board. A move consists of choosing two numbers a and b on the board, erasing them, and writing the single number

$$\frac{ab}{a+b}$$

in their place. After 19 moves exactly one number remains. Which of the following is true?

A The final number is always $\frac{1}{20}$.
B The final number is always $\frac{1}{21}$.
C The final number is always $\frac{1}{210}$.
D The final number is always 20.
E The final number is always 210.
F The final number depends on the order in which the moves are made.

Question 2

Let $N = 2^{2026}$. Consider:

I $N - 1$ is divisible by 3.
II $N + 1$ is divisible by 5.
III $N - 1$ is divisible by 7.

Which of these statements are true?

A I only
B II only
C I and II only
D I and III only
E II and III only
F I, II and III

Question 3

How many positive integers less than 1000 have the property that their digits, read from left to right, are strictly increasing? (For example 137 has this property; 370, 33 and 71 do not.)

A 84
B 120
C 129
D 130
E 165
F 512

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Question 4

Let a, b, c be real numbers. Exactly two of the following three statements are true:

$$a < b, \qquad b < c, \qquad a < c.$$

Which of the following must be true?

A $a < b$
B $b < c$
C $a < c$
D b is the largest of the three numbers
E a is the smallest of the three numbers

Question 5

For which positive integers n is it possible to fill an $n \times n$ grid with integers so that the sum of the entries in every row is odd and the sum of the entries in every column is even?

A Every $n \geq 1$
B Even n only
C Odd n only
D No n
E Every $n \geq 2$

Question 6

Let f be a continuous function on $[0, 2]$ with $\int_0^2 f(x)\,dx = 0$. Consider:

I There is some c with $0 < c < 2$ and $\int_0^c f(x)\,dx = 0$.
II There is some c with $0 < c < 2$ and $f(c) = 0$.
III There is some c with $0 \leq c \leq 1$ and $\int_c^{c+1} f(x)\,dx = 0$.

Which of these statements must be true?

A I only
B II only
C III only
D I and II only
E II and III only
F I, II and III

Question 7

What is the smallest positive integer n for which $n!$ is divisible by 10^{10}?

A 40
B 45
C 50
D 55
E 60

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Question 8

What is the smallest positive integer n with the following property: among any n integers there are two whose sum or whose difference is divisible by 12?

A 6
B 7
C 8
D 12
E 13
F 24

Question 9

How many ordered triples (x, y, z) of integers satisfy $x^2 + y^2 + z^2 = 2023$?

A 0
B 8
C 24
D 48
E 96
F Infinitely many

Question 10

A, B, C, D are four points in the plane, no three of which are collinear, with $AB = CD$ and $AD = BC$. Consider:

I $ABCD$ is a parallelogram.
II The segments AC and BD bisect each other.
III Triangles ABC and CDA are congruent.

Which of these statements must be true?

A I only
B II only
C III only
D I and II only
E I and III only
F I, II and III

Question 11

For a convex quadrilateral, the condition

"the midpoints of the four sides are the vertices of a square"

is _____ for the quadrilateral to be a square.

A necessary but not sufficient
B sufficient but not necessary
C necessary and sufficient
D neither necessary nor sufficient

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Question 12

A sequence is defined by $a_1 = 4$ and $a_{n+1} = a_n^2 - 2$ for $n \geq 1$. Consider:

I No term of the sequence is divisible by 3.
II For every $n \geq 2$, a_n leaves remainder 2 on division by 12.
III Every term of the sequence ends in the digit 4.
IV Some term of the sequence is divisible by 5.

How many of these statements are true?

A 0
B 1
C 2
D 3
E 4

Question 13

Let f be a continuous function on $[0, 1]$ with $0 \leq f(x) \leq 1$ for every x in $[0, 1]$. Consider:

I $\displaystyle\int_0^1 f(x)^2\,dx \leq \int_0^1 f(x)\,dx$
II $\displaystyle\int_0^1 f(x)^2\,dx \leq \left(\int_0^1 f(x)\,dx\right)^2$
III If $\displaystyle\int_0^1 f(x)^2\,dx = \int_0^1 f(x)\,dx$, then f is constant on $[0, 1]$.

Which of these statements must be true?

A I only
B II only
C III only
D I and II only
E I and III only
F I, II and III

Question 14

Let A, B, C be the angles of a triangle. Consider:

I $\sin A + \sin B > \sin C$
II $\cos A + \cos B > 0$
III $\sin A \sin B > \cos C$

Which of these statements must be true for every triangle?

A I only
B II only
C I and II only
D I and III only
E II and III only
F I, II and III

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Question 15

Each cell of a 6×6 grid is coloured black or white. Consider:

I Every row contains more black cells than white cells, and every column contains more white cells than black cells.

II Every row contains an odd number of black cells, and every column contains an odd number of black cells.

III Every 2×2 block of four adjacent cells contains exactly one black cell.

Which of these statements can be true?

A I only
B II only
C III only
D I and II only
E II and III only
F I, II and III

Question 16

A student is asked to prove the following claim: *if a and b are positive integers and $a^2 + b^2$ is divisible by 3, then both a and b are divisible by 3.* The student writes:

"Every square leaves remainder 0 or 1 on division by 3. If neither a nor b were divisible by 3, then $a^2 + b^2$ would leave remainder $1 + 1 = 2$, so it would not be divisible by 3. Hence at least one of a, b is divisible by 3, and the claim is proved."

Which of the following is correct?

A The claim is false: $a = 1$, $b = 2$ is a counterexample.
B The claim is true and the argument proves it.
C The claim is true, but the argument proves only that at least one of a, b is divisible by 3; the case in which exactly one of them is divisible by 3 still has to be ruled out.
D The claim is true, but the argument is invalid because a square can leave remainder 2 on division by 3.
E The claim is false: $a = b = 3$ is a counterexample.

Question 17

A finite set S of points in the plane has the property that every point of S is the midpoint of the segment joining some two other points of S. Which of the following is true?

A S must be empty.
B S can be non-empty, and any such S has at least 3 points.
C S can be non-empty, and any such S has at least 4 points.
D The four vertices of a parallelogram form such a set S.
E There is such a set S with exactly 5 points.

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Question 18

Let f be a continuous function on $[0,1]$ satisfying

$$\int_0^1 f(x)\,dx = 0 \qquad \text{and} \qquad \int_0^1 xf(x)\,dx = 0.$$

Consider:

I $f(c) = 0$ for at least one c in $(0,1)$.
II $f(c) = 0$ for at least two distinct values of c in $(0,1)$.
III $f(x) = 0$ for every x in $[0,1]$.

Which of these statements must be true?

A I only
B II only
C I and II only
D I and III only
E II and III only
F I, II and III

Question 19

The positive integer n is such that $3n+1$ and $4n+1$ are both perfect squares. What is the largest integer that must divide n?

A 4
B 8
C 14
D 28
E 56
F 112

Question 20

A pile contains 2026 stones. Two players move alternately; a move consists of removing either 2 or 3 stones from the pile. A player who cannot move loses. With best play by both sides, which of the following is true?

A The second player wins.
B The first player wins, and removing 2 stones is their only winning first move.
C The first player wins, and removing 3 stones is their only winning first move.
D The first player wins, and removing either 2 or 3 stones is a winning first move.
E The result depends on choices made later in the game.

END OF PAPER

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